

RAHUL S TAWARAKHED

Artificial Intelligence & Machine Learning | The Oxford College of Engineering, Bengaluru

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PROFESSIONAL SUMMARY

Results-driven Python Developer and final-year B.E. (AI & ML) student at The Oxford College of Engineering, with hands-on experience in machine learning, web development, and IoT systems. Proficient in Python, Flask, Streamlit, and Scikit-learn — passionate about building intelligent, real-world applications.

EDUCATION

The Oxford College of Engineering, Bengaluru <i>B.E. – Artificial Intelligence and Machine Learning CGPA: 8.604 (up to 5th Sem)</i>	2023 – Present
SMT. Pusha Mahalingappa PU College <i>Class XII (PUC) – PCMB</i>	2021 – 2023

SKILLS

Languages: Python, HTML, CSS, JavaScript, PHP

Frameworks: Flask, Streamlit,

Developer Tools: Git, GitHub, Visual Studio Code, Tinkercad

Libraries: Pandas, NumPy, Scikit-Learn

Databases: SQLite, MySQL, SQLAlchemy

IoT & Embedded Systems: Arduino, Tinkercad Simulation, Sensor Interfacing

Languages Known: Kannada, Hindi, English

PROJECTS

GreenGrow AI – Fertilizer Recommendation System [\[Github-Repo\]](#) [\[Live-Demo\]](#)

- ML-based web app that recommends suitable fertilizers based on soil nutrients (N, P, K) and environmental data; deployed for real-time predictions & 88% prediction accuracy across 6 fertilizer categories.
- Integrated Flask for backend processing and built a simple, user-friendly frontend interface.
- Trained and deployed an ML model for real-time fertilizer prediction and recommendation.

Tech Stack: Python, Streamlit, HTML, CSS, Scikit-learn, Pandas, NumPy, SQLite, Git/GitHub

AI_Based_OS – Intelligent Voice-Controlled Operating System

- Developed an AI-powered desktop operating system that enables users to open applications, browsers, files, and system utilities using text and voice commands.
- Implemented speech recognition and intelligent command processing with auto-correction to understand user intent and execute tasks efficiently & supports 25+ voice commands.
- Built a dynamic system monitoring and optimization module to track CPU, RAM, Disk, and Battery usage while maintaining low resource consumption.

Tech Stack: Python, Tkinter, SpeechRecognition, PyAudio, SQLite, psutil, Threading, OS Automation, Git/GitHub

Home Made Products – E-Commerce Web Application [\[Github-Repo\]](#) [\[Live-Demo\]](#)

- Developed a web-based platform for browsing and managing homemade products with a clean and intuitive user interface.
- Implemented product listing, product details, and user-friendly navigation to improve customer experience.
- Designed a scalable architecture that supports product management, and order processing features.

Tech Stack: Python, Render, HTML, CSS, JavaScript

EXPERIENCE

Web Development Intern – Skill Craft Technology [August 2025 – Sept 2025]

- Built 4 web projects during the internship — a Calculator, Tic-Tac-Toe (X-O) game, Responsive Portfolio website, and a To-Do web app — using HTML, CSS, and JavaScript.

Smart India Hackathon (SIH) – Participant

- Participated in India's largest hackathon, developing innovative technology solutions under tight deadlines.

CERTIFICATES

- Internship Completion Certificate – Skill Craft Technology - (2025)
- Python for Data Science – NPTEL Course Completion Certificate - (2025)

ACHIEVEMENTS

- Participated in Sentinel Hackathon - (2025)
- Participated in Smart India Hackathon (SIH) - (2025)
- Participated in SAP HackFest - (2025)